MPI LANCHESTER–OSIPOV

Verifiable Force

Quantitative assessment of the military power of states based on the Lanchester–Osipov model, with laws of combat measured on historical data and open validation of forecasts

Document 9 · Dissertation abstract

Proposed academic specialities: 1.2.2 “Mathematical modelling, numerical methods and software systems”; 2.3.1 “Systems analysis, control and information processing, statistics”; 5.5.4 “International relations, global and regional studies”

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For the executive and the investor: a one-page summary

What has been done. An open military power index of states (MPI) and a software suite for it have been created. It comprises a rating of 40 states across 253 indicators with uncertainty intervals, a combat model, a coalition builder, a calculator of missile and drone strikes against air defence, and a flashpoint monitor. The main difference from the market: **every claim of the model has been tested against real battles and wars under a protocol fixed before computation, and misses are published**.

Question	Answer
What task it solves	Comparison of the military power of states and coalitions for a specific theatre and horizon, with an honest measure of uncertainty rather than a single rating figure
Why now	World military expenditure in 2025 — 2.887 trillion $ (a SIPRI record), the wars of 2022–2026 have refuted counting ratings; demand for testable models is growing, while the niche leader TENET has moved from an index to an AI platform for analysts
How it is better than competitors	Open formulas and code; combat laws measured on 404 battles; validation on 10 wars of 2018–2025; explicit model error; a hot-spot monitor with AUC 0.96 out of sample (a simple baseline from border, history and last year gives 0.958 — we acknowledge this); competitors (Global Firepower, TENET MCI) disclose neither formulas nor validation
What already works	A Python calculation core (more than 60 automated tests, a reference ranking test; the code is closed until version 1.0, available to reviewers on request), a browser interface prototype, 9 Word reports, data collection from 30+ open sources, the Delphi expert panel protocol
What is missing so far	Coverage — 40 states against 145–170 for competitors; IISS data are not licensed; 67 “quality” indicators are temporarily filled with an AI estimate pending the expert panel; there is no sales team and no clients
Revenue model (options)	Platform subscription; a licence for the data and the methodology; commissioned scenarios and reports; training (military academies, think tanks)
The main risk	The model so far predicts war outcomes no better than a uniform forecast — this is acknowledged and published; the value of the product lies in an honest balance of power and in scenarios, not in “predicting wars”

> **Position.** unlike the market, we do not promise accurate forecasts of war outcomes: our own validation has shown that open data do not permit this. We sell testability — what competitors promise but keep closed.

1. General description of the work

Relevance of the topic

Assessment of the military power of states is needed for military planning, foreign policy, the defence industry, insurance and investors. Widely used ratings (Global Firepower, CINC) add up the quantity of equipment and resources linearly, with opaque or arbitrary weights, without accounting for quality, readiness, geography, time, allies or uncertainty. The wars of 2022–2026 showed the cost of these errors: in the 2022 GFP rating Russia stood in 2nd place and Ukraine in 22nd, yet the war became a protracted positional one; Israel and Iran stood close to each other in 2025, yet Israeli aviation achieved air supremacy over Iran within days.

The classical Lanchester–Osipov model (1915–1916) sets the form of the relationship between numbers and quality, but its laws are poorly supported by history, and when transferred to the level of states it loses economics, geography, alliances and the nuclear factor. A methodology is needed that links the index and the combat model and **tests itself against data**.

Degree of elaboration of the problem

Combat models evolved from the laws of Lanchester and Osipov to the modifications of Helmbold (1965) and Deitchman (1962), the Hughes salvo model (1995), three-sided models (Kress et al., 2018) and spatial models (Randow et al., 2026). Fitting to battles (Engel, 1954; Bracken, 1995; Fricker, 1998; Lucas, Turkes, 2004) produced contradictory results. Power indices (CINC, Souva MMP, DOE Carroll–Kenkel, Beckley) developed separately from combat models. Commercial ratings (Global Firepower, TENET MCI) are closed and do not publish validations. The chain “index → combat model → out-of-sample validation under a fixed protocol” is absent from the literature reviewed.

Aim and objectives

Aim — to create an open, testable methodology and software suite for the quantitative assessment of the military power of states and coalitions, linking the structural index with the combat model.

- To analyse combat models and power indices and identify their systematic errors.
- To build an index of 253 indicators with roles in the model, non-linear aggregation, a nuclear axis, power projection and uncertainty intervals.
- To measure the laws of combat (law exponent, intensity, quality factors, will threshold) on historical battles.
- To validate the model on the wars of 2018–2025 using data from the year before the war and to estimate its error.
- To develop models of coalitions, drone and missile strikes against air defence, and a flashpoint monitor.
- Create a software suite with automated tests and an interface; the methodology, formulas and parameters are open, the core code is closed until version 1.0.

Object and subject of research

Object of research — the military power of states and coalitions. **Subject of research** — mathematical models and methods for its quantitative assessment and their validation against historical data.

Scientific novelty

- **The logarithmic form of Lanchester's law as the rule for index weights:** $\ln F = \ln Q + \alpha \cdot \ln N$, where α is measured on battles rather than taken from a textbook; each of the 253 indicators is assigned a role in the combat equations.
- **Measurement of the laws of combat on 404 battles of 1900–1973 (CDB90) under protocols fixed before computation:** the quadratic law is rejected (α = 0.65 [0.46; 0.84]; adjusted for combat effectiveness 0.98 [0.80; 1.18]); the form p + q = 1 is rejected for combat intensity (p + q = 0.62 [0.55; 0.70]).
- **Quality factors in units of numerical advantage:** one step of superiority in leadership equals an advantage of ≈ 9 times, in morale ≈ 6, in supply ≈ 2.3, in intelligence ≈ 2.2 (upper estimates: the codes were assigned after the events).
- **The will threshold with censoring:** the median losses at which a side stops fighting is 16% [14; 18]; one step of morale raises the threshold by a factor of 1.29. An independent survey (Podobnik et al., 2025) gives a median acceptable loss of 10%.
- **The defence coefficient of a specific war based on its loss ratio** (Russia against Ukraine — 1.7 according to Oryx, 2024–2025) and **an explicit model error σ = 2.0**, calibrated on 10 wars.
- **The first, as far as is known, out-of-sample test of the hypothesis that "the depth of production decides the drone war"** — the hypothesis is not supported.
- **An open log of negative results:** calibration, year-by-year quality, the strike model, drones — everything that did not improve the forecast is published with the reasons.

Theoretical and practical significance

Theoretically, the work refines the scope of applicability of Lanchester's laws (a nearly linear law for the loss-exchange ratio, incomplete force participation for intensity) and provides measurable quality multipliers. Practically, it offers a software suite for comparing states and coalitions, war scenarios, air-defence assessment and flashpoint monitoring; the methodology is applicable in military academies, think tanks, the defence industry and political risk insurance.

Methodology and methods

Systems analysis; differential equations of combat; CES function; Monte Carlo method; regression and censored models (log-normal threshold model); evaluation of probabilistic forecasts by the logarithmic score (Gneiting, Raftery, 2007); leave-one-out cross-validation; Sobol sensitivity indices; Delphi method with Cooke weights; logistic regression for conflict forecasting; open-source data processing. **Working rule:** the protocol of every validation is fixed by a git commit before computation; the protocol cannot be changed after computation.

Propositions submitted for defence

- The military power index, in which indicators enter the formulas according to their roles in the combat equations and the pillars are aggregated with a CES function, eliminates the systematic errors of counting ratings: compensation of the "weakest link", summing nuclear weapons with conventional forces, ignoring distance and uncertainty.
- On the battles of 1900–1973 the loss-exchange ratio is described by a nearly linear law (α ≈ 1 adjusted for effectiveness), the quadratic law is rejected; combat intensity grows more slowly than numbers (p + q ≈ 0.62).
- The quality of troops (leadership, morale, supply, intelligence) is equivalent in strength to a several-fold numerical advantage and must enter the index as multipliers, not as addends.

- A battle ends with a collapse of will at median losses of about 16%, not with annihilation; the threshold is a scenario parameter (battle — 16%, campaign — 40%).
- The balance of power by the index from open data does not predict war outcomes better than a uniform forecast; an honest model must include an explicit error ($\sigma = 2.0$) and publish this.
- The probability of armed conflict in a pair of states is determined by proximity, history and institutions (AUC 0.96 out of sample; almost as much, 0.958, is given by a simple baseline from border, history and last year), not by the balance of military power.
- Open country-level features (GDP per capita, democracy, mechanisation) explain a fifth of the loss exchange in the battles of 1900–1973 but do not transfer to interstate clashes of 1989–2024 (UCDP GED: Spearman +0.25 with an interval from −0.60 to +0.63, 24 cases).

Reliability and approbation

Reliability is ensured by open formulas, parameters and protocols and by data with a hash manifest (the core code is closed until version 1.0, available to reviewers on request); more than 60 automated tests (checks of formulas, the battle model, the panel and calibration) run in GitHub Actions; a check of the browser calculation against the core calculation; protocols fixed before the calculation (commits 567224b, 65e1a59, 5ed015a, f2a1a3c, 67504ee, cb72541, 061ed46 and others); and publication of negative results. The materials are presented in 9 reports (documents 1–9) and an interactive prototype.

Structure of the work

The work consists of an introduction, eight chapters (review and criticism of models; the index; laws of combat on battles; validation against wars; drones and air defence; coalitions; the flashpoint monitor; the expert panel and the software suite), a conclusion, a bibliography and appendices (the catalogue of 253 indicators, validation protocols, data).

2. Main content of the work

Chapter 1. Combat models and power indices: a critical review

The Lanchester and Osipov laws (linear, quadratic, logarithmic), the modifications of Helmbold, Deitchman and Hughes, three-sided combat, and 13 indices and methods of power assessment are examined (Global Firepower, CINC, Lowy, BICC, Heritage, Souva, DOE, Beckley, Credit Suisse, Dupuy's QJM/TNDM, WEI/WUV and the “potential–antipotential” method, the domestic method of combat potentials). Common errors: counting “units” without quality and readiness, double-counting the budget and purchased equipment, summing nuclear weapons with conventional forces, linear aggregation, untested weights, point estimates without intervals, ignoring geography, time and allies.

Chapter 2. Military power index: construction

253 indicators are grouped into 12 pillars (ground forces, air force and air defence, navy, strategic forces, space and command, personnel, economy, defence industry, logistics, geography, alliances, governance and experience). Each indicator is assigned a role: N — numbers (41), Q — quality (57), M — multiplier (35), S — sustainment (81), P — projection (13), G — deterrence (11), C — context (15).

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(1) Normalisation of count quantities:  x̃ = lg(x) on a scale three orders of magnitude
below the sample leader
(2) Combat pillar (land, air, sea):
    P_d = (α·Σw·x̃[N] + Σw·x̃[Q,M] + Σw·x̃[other]) / (α·Σw[N] + Σw[Q,M] + Σw[other])
    — the logarithmic form of Lanchester's law  ln F = ln Q + α·ln N
(3) Pillar aggregation (CES):  MPI = (Σ ω_p · P_p^ρ)^(1/ρ),  ρ = 0.5
    ρ = 1 — sum; ρ → 0 — geometric mean; ρ < 0 — “weakest link”
(4) Nuclear axis: class 0–4 outside the aggregation; modes “axis”, “pillar”,
“multiplier” (1 + 0.3·P_STR)/1.3
(5) Power projection (Boulding):  MPI(D) = MPI · e^(–D/λ),  λ = λ₀ + λ₁·P̄[role P]
(6) Conversion into c.u. of power:  M = exp(κ·MPI/100),  κ = 2.6·ln 10 ≈ 6.0
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Uncertainty: each observation is assigned a reliability class (A–D) and an error σ; the Monte Carlo method (1,000 runs) varies the data, the weights (× LogN(0; 0.15²)), ρ ∈ [0.3; 0.7] and α ∈ [0.8; 1.2]; the result is the MPI interval and the range of ranks. Pilot: 40 states (20 major powers and 20 flashpoint participants), 215 indicators in the index, the snapshot is 83.9% complete (without the temporary AI estimate — 57.4%). Result: United States 79.6 [77.2; 80.0], China 75.5, Russia 70.4; the rank correlation with the expert assessment of 8 countries — 0.81. Sensitivity (Sobol): model structure — 0.36 of the spread, data — 0.34.

Chapter 3. Laws of combat on historical battles

The material is the US Army's CDB90 database: 404 battles of 1900–1973 with numbers, losses, duration and the compilers' expert factors. All calculations follow protocols fixed before computation.

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(7) Generalised combat model:  dA/dt = –a · A^(1–α/2) · B^(α/2) + r_A(t)
                               dB/dt = –b · B^(1–α/2) · A^(α/2) + r_B(t)
    invariant without reinforcements:  b·A^α – a·B^α = const
(8) Required force ratio:  R = (e_B · d / e_A)^(1/α)
(9) Combat intensity:  ln(L/T) = c + p·ln N_enemy + q·ln N_own
(10) Factor multiplier in units of advantage:  eq = exp(β_factor / β_ln(N_A/N_B))
(11) Will threshold (censored log-normal model):  ln m* ~ N(μ, s²),
     observed losses f — the lower bound of the threshold for the side that holds
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Result	Value	Conclusion
Law exponent α (under the protocol)	0.65 [0.46; 0.84]	the square law (α = 2) rejected
α adjusted for effectiveness	0.98 [0.80; 1.18]	a nearly linear law

Result	Value	Conclusion
Intensity: p, q, p + q	0.16; 0.46; 0.62 [0.55; 0.70]	doubling of both sides → daily losses × 1.54, not × 2: only part of the forces takes part in battle
Leadership / morale / supply / intelligence	≈ 9 / 6 / 2.3 / 2.2 times the advantage	quality matters more than numbers (upper estimates)
Attackers win	68% of battles	the force ratio hardly improves the forecast
Will threshold m*, median	16% [14; 18]	combat ends with a collapse of will
Morale step	threshold × 1.29 [1.12; 1.48]	morale raises staying power

Chapter 4. Validation against wars and model error

Calibration against 5 wars of 2020–2026 (1,350 parameter combinations, cross-validation): the parameters that are best in the sample are worse than the defaults on held-out wars (−31.7 versus −19.6) — the defaults were retained; there is no common κ at any α. **Extended validation against 10 wars of 2018–2025** using the UCDP register, with the index based on data from the year before the war (72 identical indicators): all model variants are worse than the uniform forecast (−14.6); variant θC was adopted (α = 1.0; d = 1.3 / 1.0 / 1.0). **Model error:** σ = 2.0 in the probability that the advantage is sufficient:

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(12) P(the advantage is sufficient) = Φ( ln(r ÷ R) / σ ),  σ = 2.0
(13) Logarithmic score of the forecast:  S = Σ ln p(observed outcome)
```

Defence coefficient of the Russia–Ukraine war based on Oryx equipment losses in 2024–2025 — d = 1.7. The scenario “Russia against Ukraine” yields a ratio of 1.68 : 1 against the required 1.70 : 1 — a positional war, which matches reality (but is not an independent validation). **Year-by-year quality** (expenditure per service member, education) and **a separate model of strike campaigns** did not improve the forecast — the results are published.

Chapter 5. Drones, salvoes and air defence

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(14) Hughes salvo model:  ΔB = (a·A − y_B·B) / w_B,   ΔA = (b·B − y_A·A) / w_A
(15) Air-defence saturation:  leakage share = (1−f)·(1−p)^n + f·(1−p)^(n+1),  k̄ = n + f
     (integer interceptors, uniform distribution — optimal according to arXiv
2608.31143)
(16) Drone hypothesis:  ln r_t = c + b·x_t,  x_t = ln(Ukraine's output ÷ Russia's
output)
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Testing the hypothesis “the depth of production decides the drone war” (Gogerty, 2026) under protocol 061ed46: the ratio of Russian and Ukrainian equipment losses by year — 3.20; 2.31; 2.41; 1.16. Between 2023 and 2024 Ukraine's relative output fell (x from 1.74 to 0.38), while the loss ratio did not change (b ≈ −0.03); the forecast for 2025 is 2.38 against an actual 1.16, and the direction is wrong. **The hypothesis is not supported.** The literature (RUSI, 2025; Defence Studies, 2026) points to better inputs: 20–40% of drones reach the target, and about 90% of the fleet is expended within a year.

Chapter 6. Coalitions

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(17) F_coal(t) = F_own + Σ_j ρ_j · s_j · F_j · e^(−D_j/λ_j) · g(t − τ_j)
```

ρ is ally reliability (commitment classes A–F: from a collective defence treaty to an informal axis), s is the force share in theatre, τ is the entry delay. Contribution types: troops, supply, intelligence, territory, finances. Coalition scenarios use “autonomous power” without the alliances pillar so that allies are not counted twice. Scenarios: Russia against Ukraine (1.68 : 1 against the required 1.70); Israel and the United States against Iran (6.84 : 1 against 1.30); the Baltics (the attacker's window of advantage — days 0–7, then NATO advantage).

Chapter 7. Flashpoint monitor

780 pairs from 40 states × 2016–2025, outcome — armed conflict according to UCDP, features of the previous year. The model “proximity, distance, history, conflict last year, democracy (V-Dem), alliance” — **AUC 0.96 out of sample** (5,460 forecasts, 38 conflicts; the simple baseline “border + history + conflict last year” gives 0.958, the full model’s gain of +1.28 is below the threshold of 2 — the acceptance rule is not met); onsets of new conflicts — AUC 0.75 (8 cases). Military power indicators do not improve the forecast (−68.6 versus −66.7) and are not included in the monitor. The 2026 forecast is published in the prototype.

Chapter 8. Expert panel and software suite

Delphi panel: 67 questions in 9 competence modules, 0–4 scales with observable anchors or numbers with an 80% interval; competence coefficient $K_c \geq 0.5$; exclusion of assessments of the expert's own country; consensus — interquartile range; agreement — Kendall's W; Cooke calibration on 12 seed questions. Until the panel concludes, 67 indicators are filled with a temporary AI estimate (lowest reliability class; the calculation without it is published alongside).

Software suite: data collection from 30+ open sources (Python), the calculation core `core/lanmil` (normalisation, imputation, index, Monte Carlo, battle model, coalitions, salvos, validation, sensitivity), more than 60 automated tests in GitHub Actions, a single parameter file for the core and the prototype with a self-check, an interactive prototype (ranking, profile, comparison, battle model, salvos and air defence, coalitions, hot-spot monitor, methodology, catalogue), Word and Excel report generators.

3. Conclusion

Main conclusions

- Counting ratings are systematically wrong; an index with indicator roles, non-linear aggregation, a nuclear axis, projection and intervals eliminates these errors by construction.
- The Lanchester square law is not confirmed by history: the loss-exchange ratio is nearly linear ($\alpha \approx 1$), and combat intensity grows more slowly than numbers.
- The quality of troops (leadership, morale, supply, intelligence) is equivalent in strength to a several-fold numerical advantage; a battle ends with a collapse of will at losses of about 16%.
- From open data, the balance of power does not predict war outcomes better than a uniform forecast; the model must carry an explicit error, and this must be stated to the client plainly.
- Wars are not started by the strongest: the probability of conflict is determined by geography, history and institutions.
- The simple link “drone output → loss-exchange ratio” is not confirmed; measurable inputs are needed: consumption, the share reaching the target, losses to electronic warfare (EW).

Limitations

Coverage — 40 states; no licensed IISS data; 67 quality indicators are temporarily filled with an AI estimate; there are few validation wars (5–10), which limits calibration; CDB90 factors were assigned after the events and yield upper estimates; loss data for ongoing wars are biased by the sources.

Prospects

- An open forecast log: the first entry — monthly losses of Russia and Ukraine for Q4 2026, fixed before the period begins.
- An expert Delphi panel instead of the AI estimate; expansion to 150+ states.
- An attrition model with drones on measurable inputs; layered air defence (missiles + interceptor drones).
- Adjustment of loss data for source bias (Radford et al., 2023); checking the war's d on WarSpotting.
- Platform (FastAPI, React), a public website in two languages, a data DOI and a preprint on the CDB90 battles.

4. Competitive environment in the United States

Below are the US organisations closest to the MPI in terms of task. Important players outside the US are listed separately. Data are from the organisations' websites and the press as of 13.10.2026.

Organisation	Product	Strengths	Weaknesses	How we differ
Global Firepower (Virginia, Utah)	a rating of 145 countries, 60+ factors	coverage, recognition, annual cadence	the formula is closed, there are no intervals or validation; in 2022 the counting gap between Russia and Ukraine did not materialise	open formulas, intervals, validation against wars
TENET Intelligence (Arlington)	MCI: 150+ countries, 200+ indicators; since 2026 — the SAGE platform	an advisory board (former NATO commander, former intelligence chiefs), client testimonials	the core is closed; the square law is untested; the forecast log is closed; 29 of 50 sitemap addresses — 404	the combat law is measured; validations and misses are open
Heritage Foundation	Index of U.S. Military Strength	authority, expertise	only the US armed forces, a qualitative scale	comparison of states and coalitions
Verisk (Maplecroft)	Predictive War Index (May 2026): the probability of war in a country over 12 months	a risk-modelling brand, clients — banks and insurers	machine learning without published validation; trained on 1995–2022	a flashpoint monitor with a published out-of-sample AUC of 0.96 and a protocol
Rhombus Power	Guardian: AI decision support and early warning (US Air Force)	access to the customer — the Pentagon	closedness, confined to the defence sector	an open methodology for the civilian and expert market
US academic projects (COW/CINC; Souva MMP; DOE Carroll–Kenkel)	open datasets and measures of power	peer review, long data series	no product, scenarios or interface	a product on a validated methodology

Outside the United States	Why it matters
IISS Military Balance+ (United Kingdom)	the main source of data on the armed forces; a licence is needed to expand coverage
Janes (United Kingdom)	databases of 40 thousand military units; the benchmark of the data market
GlobalMilitary.net — ATLAS Index (country not identified)	169–170 countries, published weights, a worked example and a sensitivity table — the rating closest to ours in openness, but without a combat model or validation against wars
VIEWS (Norway, Sweden), ACLED	open early-warning systems for violence; a benchmark for the flashpoint monitor
Hadean (United Kingdom)	AI combat simulations, a round of about $15 million (2025–2026), investor — Booz Allen Ventures

Conclusion for the investor. the market is divided: ratings are open but untested; defence analytics platforms are closed and expensive. The niche of an “open, testable index with a combat model” is vacant. Demand is confirmed by US Air Force requests (WarMatrix — scenarios 10,000 times faster than real time) and by insurers entering war modelling.

Bibliography and sources

A. Classical theory of combat

1. Osipov M. The influence of the numbers of the fighting sides on their losses // Military Collection. 1915. No. 6–10.
2. Lanchester F. W. Aircraft in Warfare: The Dawn of the Fourth Arm. London: Constable, 1916.
3. Engel J. H. A verification of Lanchester's law // Journal of the Operations Research Society of America. 1954. Vol. 2, № 2. P. 163–171.
4. Deitchman S. J. A Lanchester model of guerrilla warfare // Operations Research. 1962. Vol. 10, № 6. P. 818–827.
5. Helmbold R. L. A modification of Lanchester's equations // Operations Research. 1965. Vol. 13, № 5. P. 857–859.
6. Schaffer M. B. Lanchester Models of Guerrilla Engagements. RAND RM-5053-ARPA, 1967.
7. Boulding K. E. Conflict and Defense: A General Theory. New York: Harper, 1962.
8. Snyder G. H. The balance of power and the balance of terror // Seabury P. (ed.) The Balance of Power. San Francisco: Chandler, 1965.
9. Dupuy T. N. Numbers, Predictions and War. Indianapolis: Bobbs-Merrill, 1979.
10. Hughes W. P. A salvo model of warships in missile combat used to evaluate their staying power // Naval Research Logistics. 1995. Vol. 42, № 2. P. 267–289.
11. Kress M., Caulkins J. P., Feichtinger G., Grass D., Seidl A. Lanchester model for three-way combat // European Journal of Operational Research. 2018. Vol. 264, № 1. P. 46–54.
12. Coulson S. Lanchester modelling of intelligence in combat // IMA Journal of Management Mathematics. 2019. Vol. 30, № 2. P. 149.
13. Atkinson M. P., Kress M., MacKay N. J. Targeting, deployment, and loss-tolerance in Lanchester engagements // Operations Research. 2021. doi 10.1287/opre.2020.2022.
14. Shumov V. V. Taking account of psychological factors in combat (conflict) models // Computer Research and Modeling. 2016. Vol. 8, No. 6. P. 951–964.
15. Mitskevich. Methodological materials: the ratio of branches of the armed forces, the Osipov–Lanchester model. Minsk: BSUIR.
16. Shleiko. A method for assessing the effectiveness of fire engagement // KazNU Bulletin.
17. Yusupov R. M., Ivanov. Russian mathematical models of combat operations of the early 20th century // Informatics and Automation. 2023.
18. Combat capabilities // Great Russian Encyclopedia (bigenc.ru).

B. Validation of combat models on historical data

19. Bracken J. Lanchester models of the Ardennes campaign // Naval Research Logistics. 1995. Vol. 42, № 4. P. 559–577.
20. Clemens S. C. The application of Lanchester models to the Battle of Kursk. 1997.
21. Fricker R. D. Attrition models of the Ardennes campaign // Naval Research Logistics. 1998. Vol. 45, № 1. P. 1–22.
22. Hartley D. S. Predicting Combat Effects. Linthicum: INFORMS, 2001.
23. Helmbold R. L. Historical support for a mixed law Lanchestrian attrition model: Helmbold's ratio. OSTI 5087936.
24. Lucas T. W., Turkes T. Fitting Lanchester equations to the battles of Kursk and Ardennes // Naval Research Logistics. 2004. Vol. 51, № 1. P. 95–116.
25. Fitting Lanchester and Other Equations to the Battle of Kursk. DTIC ADA378391.
26. Rubio-Campillo X. Model selection in historical research using Approximate Bayesian Computation // PLoS ONE. 2016. Vol. 11, № 1. e0146491.
27. Plowes N. J. R., Adams E. S. An empirical test of Lanchester's square law: mortality during battles of the fire ant Solenopsis invicta // Proceedings of the Royal Society B. 2005. PMC1559866.

28. Concepts Analysis Agency. CDB90: Database of Battles. US Army, 1990 (jrnold/cdb90 version, ODC-By licence).
29. Chung D., Jeong B. Analyzing Russia–Ukraine war patterns based on Lanchester model using SINDy algorithm // Mathematics. 2024. Vol. 12, № 6. 851. doi 10.3390/math12060851.
30. The Least Action-Augmented Lanchester Model. arXiv 2512.15058, 2025.
31. Aggregate combat modeling using high-resolution simulation: the «meeting engagement» scenario as a case study. arXiv 2607.15312, 2026.

C. Military and national power indices

32. Singer J. D., Bremer S., Stuckey J. Capability distribution, uncertainty, and major power war, 1820–1965 // Russett B. (ed.) Peace, War, and Numbers. Beverly Hills: Sage, 1972. (CINC index, Correlates of War project)
33. Carroll R. J., Kenkel B. Prediction, proxies, and power // American Journal of Political Science. 2019. Vol. 63, № 3. P. 577–593.
34. Beckley M. The power of nations: measuring what matters // International Security. 2018. Vol. 43, № 2. P. 7–44.
35. Souva M. Material military power: a country-year measure of military power, 1865–2019 // Journal of Peace Research. 2023. Vol. 60, № 6. P. 1002–1009.
36. Global Firepower. 2026 Military Strength Ranking (globalfirepower.com).
37. Lowy Institute. Asia Power Index 2025: Key Findings.
38. BICC. Global Militarization Index 2025.
39. The Heritage Foundation. 2026 Index of U.S. Military Strength.
40. Credit Suisse Research Institute. The End of Globalization or a more Multipolar World? 2015.
41. GlobalMilitary.net. ATLAS Index: Military Rankings Methodology (2026).
42. Saaty T. L. The Analytic Hierarchy Process. New York: McGraw-Hill, 1980.
43. Antipotential-potential method and weapon scores. DTIC ADA295844.
44. The Dupuy Institute. Scoring weapons and aggregation in Trevor Dupuy's combat models. 2018.
45. Ben-Haim Y. WEI–WUV and info-gap robustness. Technion.
46. Vox Ukraine. Global Firepower — a «military ranking» without verified facts or transparent methodology. 2025.

D. Drones, air defence and modern war

47. Watling J., Reynolds N. Tactical Developments During the Third Year of the Russo-Ukrainian War. London: RUSI, February 2025.
48. Plichta M. Precise mass in action: assessing Ukraine's one-way attack drone campaign // The RUSI Journal. 2025. Vol. 170, № 4. P. 42–48.
49. Berthelsen E. Innovating at the pace of relevance: the case of Ukrainian unmanned systems // Defence Studies. 2026. doi 10.1080/14702436.2026.2692416.
50. Randow C., Allaire R. H., Mastroberardino A., Luo R. Modeling drone swarms with a PDE extension of Lanchester's laws // Journal of Defense Modeling and Simulation. 2026. doi 10.1177/15485129261436073.
51. Gogerty N. From Square Law to Swarm Law: A Drone-extended Lanchester Model and the New Mathematics of Attrition across Eight Conflicts. SSRN 6881358, 2026.
52. Pokorny L. The Economics of Asymmetric Attrition: A Quantitative Analysis of Low-Cost Drone Warfare in the Ukraine and Iranian Shahed Programs (2022–2026). Zenodo, 2026. doi 10.5281/zenodo.18919501.
53. On the problem of assigning multiple interceptors over multiple aerial threats. arXiv 2608.31143, 2026.
54. Scaling and Trade-offs in Multi-agent Autonomous Systems. arXiv 2603.10743, 2026.
55. Modeling Large-Scale Adversarial Swarm Engagements using Optimal Control. arXiv 2602.23323, 2026.

56. Forecasting Russian Equipment Losses Using Time Series and Deep Learning Models. arXiv 2509.07813, 2025.
57. Radford B. J., Dai Y., Stoehr N., Schein A., Fernández M., Sajid H. Estimating conflict losses and reporting biases // PNAS. 2023. Vol. 120, № 34. e2307372120.
58. Podobnik B., Medvidovic L., Wild D., Boccaletti S. Empirical study and model of warfare dynamics: willingness to risk // Chaos. 2025. doi 10.1063/5.0277615.

E. Statistical and expert methods

59. Dalkey N., Helmer O. An experimental application of the Delphi method to the use of experts // Management Science. 1963. Vol. 9, № 3. P. 458–467.
60. Kendall M. G., Babington Smith B. The problem of m rankings // Annals of Mathematical Statistics. 1939. Vol. 10, № 3. P. 275–287.
61. Cooke R. M. Experts in Uncertainty: Opinion and Subjective Probability in Science. Oxford University Press, 1991.
62. Sobol' I. M. Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates // Mathematics and Computers in Simulation. 2001. Vol. 55. P. 271–280.
63. Saltelli A. et al. Global Sensitivity Analysis: The Primer. Chichester: Wiley, 2008.
64. Gneiting T., Raftery A. E. Strictly proper scoring rules, prediction, and estimation // Journal of the American Statistical Association. 2007. Vol. 102, № 477. P. 359–378.

F. Data sources

65. SIPRI Military Expenditure Database; SIPRI Fact Sheet “Trends in World Military Expenditure, 2025” (April 2026); SIPRI Top-100 (December 2025); SIPRI Yearbook 2026, ch. 8 “World nuclear forces”; SIPRI Arms Transfers Database (October 2026).
66. IISS. The Military Balance 2024–2026.
67. World Bank: World Development Indicators, Worldwide Governance Indicators, Human Capital Index, Data360, Climate Change Knowledge Portal (CRU TS 4.08).
68. IMF. World Economic Outlook (DataMapper).
69. Our World in Data (Energy Institute Statistical Review); UN World Population Prospects 2024.
70. CIA World Factbook; UCS Satellite Database (May 2023).
71. Gleditsch N. P. et al. Armed conflict 1946–2001: a new dataset // Journal of Peace Research. 2002. Vol. 39, No. 5. P. 615–637; UCDP/PRIO Armed Conflict Dataset, version 26.1.
72. Leeds B. A. et al. Alliance Treaty Obligations and Provisions, 1815–1944 // International Interactions. 2002. Vol. 28, № 3. P. 237–260; ATOP 5.1.
73. Coppedge M. et al. V-Dem Dataset; Teorell J. et al. Quality of Government Standard Dataset (January 2026); World Values Survey / European Values Study.
74. Bjørnskov C., Rode M. Regime types and regime change: a new dataset on democracy, coups, and political institutions // Review of International Organizations. 2020. Vol. 15. P. 531–551.
75. Nunn N., Puga D. Ruggedness: the blessing of bad geography in Africa // Review of Economics and Statistics. 2012. Vol. 94, № 1. P. 20–36.
76. Transparency International CPI 2025; WIPO Global Innovation Index 2025; worldsteel; FAS Nuclear Notebook; ITU Global Cybersecurity Index 2024.
77. UN — Uniformed Personnel Contributions; USGS Mineral Commodity Summaries 2026; FAOSTAT; Secure World Foundation “Global Counterspace Capabilities” (2025); CSIS Missile Threat.
78. Natural Earth; TeleGeography Submarine Cable Map; U.S. EIA «World Oil Transit Chokepoints»; OurAirports; UNCTADstat; UNESCO UIS; CSET «The Global Distribution of STEM Graduates» (2023); UNIDO.
79. Oryx. Documenting Russian/Ukrainian Equipment Losses (Internet Archive Wayback Machine snapshots, 2022–2026); WarSpotting.
80. Wikipedia: tables of the armaments of the armed forces and a list of countries with overseas bases (IISS, GFP, Oryx data in the notes).

G. Chronicle of events and the competitive environment

81. Britannica “2026 Iran war”; CFR Global Conflict Tracker; International Crisis Group; ISW; Reuters, AP, Al Jazeera, PBS, Fortune, Financial Times — chronicle of 2025–2026.
82. TENET Intelligence: A Quantitative Framework for Modern Military Power: The TENET MCI; Force Size vs. Force Quality; Forecast Record (tenetintel.com, 2025–2026; copies — Wayback Machine).
83. Verisk Maplecroft. Predictive War Index (May 2026) // Fortune. 14.06.2026.
84. Rhombus Power. Guardian (rhombuspower.com; Federal News Network, 2023).
85. Johns Hopkins APL. GenWar Lab (2025); U.S. Air Force RFI «WarMatrix Ecosystem» (2025).
86. VIEWS — Violence & Impacts Early-Warning System (PRIO, Uppsala University); ACLED Conflict Index 2026.

Total sources: 86. References to pages and datasets — in documents 1–8 and in the project's docs/*.md.